

Kamalesh Reddi Arugunta

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Professional Summary

Software Engineer candidate focused on distributed systems, artificial intelligence (AI), and machine learning (ML), with experience across the software development life cycle (SDLC): RESTful APIs, AI agents, full stack web, cloud infrastructure, and DevOps. Co-founder of Tac LLC, where team brainstorming and a growth mindset turned a creative idea into a supportive Braille device for visually impaired students. Inquisitive, consistent, and collaborative, with a strong foundation in software design and product sense.

Education

Arizona State University Tempe, AZ
Master of Science in Computer Science, Data Science Concentration | **GPA: 4.0/4.0** Jan 2026 – Dec 2026

Arizona State University Tempe, AZ
Bachelor of Science in Computer Science | **GPA: 4.0/4.0** Aug 2022 – Dec 2025

- **New American University Scholar** | **Accelerated Bachelor's/Master's (BS/MS) Program**
- **Relevant Coursework:** Discrete Mathematics, Linear Algebra, Probability & Statistics, Data Structures, Algorithms
- **Teaching Assistant (TA) / Grader:** Supportive mentorship to 150+ students in collaboration with instructors; taught C# and .NET

Skills

Programming Languages: Python, C/C++ (C++ programming), Java, Go, JavaScript, TypeScript, SQL, R, C#
AI/ML: Machine Learning (ML), Deep Learning, PyTorch, CUDA, Large Language Models (LLMs), LangGraph, Retrieval-Augmented Generation (RAG)
Data: Apache Spark (Spark), Apache Kafka (Kafka), Extract, Transform, Load (ETL), Data Quality, PostgreSQL, CockroachDB
Web / Full Stack: React, HTML, CSS, D3.js, Flask, SQLAlchemy, MongoDB
Backend & Cloud: Distributed Systems, Amazon Web Services (AWS) EC2, S3, Lambda, DynamoDB, Microsoft Azure, Google Cloud Platform (GCP), Linux
DevOps & Tools: DevOps, Docker, Kubernetes, Continuous Integration/Continuous Delivery (CI/CD), Jenkins, GitHub Actions, Git, GitHub, GitLab
Testing & Debugging: Benchmarking, Debugging, Triage, Root-Cause Analysis, Pytest, Prometheus, Grafana
Fundamentals & Practices: Algorithms, Software Design, Product Sense, Object-Oriented Programming (OOP), SDLC

Certifications

Amazon Web Services (AWS) Certified Cloud Practitioner | Microsoft Certified: Azure Fundamentals (AZ-900)

Professional Experience

Samtek Inc. May 2026 – Jul 2026
Software Engineer Intern Herndon, VA

- Extended endpoint security to **150 endpoints** by building **REST API** integrations and providing application support
- Guided a **150-device** migration via requirements analysis of **Trend Micro vs. Defender** and **JumpCloud vs. Intune**
- De-risked the **150-device** rollout via a **pilot migration playbook**, technical documentation, and technical specifications

Arizona State University Jan 2026 – Present
Graduate Student Researcher Tempe, AZ

- Surfaced **3 critical gaps** in privacy curriculum by surveying **1,000+ U.S. faculty** in a collaborative research team
- Turned a **1,000+ contact dataset** into interactive **D3.js dashboards** of trends, normalized via **Python (pandas)**

Ada Analytics May 2025 – Jan 2026
Machine Learning Intern Remote

- Generated **10%+ backtested profit margins** on automated equity selection by training **PyTorch Deep Learning sentiment models** on congressional trading and 13F filings, surfacing top picks via a full stack **React/shadCN** dashboard
- Scored **1,000+ equities** daily via a **GCP-hosted Python ETL and data processing pipeline** with data quality checks on scraped market and alternative-data APIs, ranking the top 10 trades using a **3-signal AI/ML classification model**
- Automated execution of the **top 10 daily trades** by wiring **AI/ML** signals to the **Alpaca REST API** with risk gating

Projects

Multi-Pattern Matching Engine for Network Intrusion Detection (C++20, AVX2 SIMD, Linux) Sep 2026

- Reached **6,551 MiB/s** (5.3x over scalar) by building a streaming **Aho-Corasick** engine with an **AVX2** prefilter and lock-free ring
- Validated correctness on **1,000,000** differential test cases by automating GCC/Clang benchmarks, ASan/UBSan, and fuzzing in CI

CineSync - Distributed Event Ticketing Platform (Python, Flask, SQLAlchemy, CockroachDB) Dec 2025

- Kept **100% booking integrity** under node failures (**full credit**, CSE 512) by replicating data across a **3-node CockroachDB** cluster
- Guaranteed **zero double-bookings** by designing an **8-entity schema** with composite keys and **PostgreSQL-compatible transactions**

ResQ-Agents - Disaster Response Orchestration Platform (Python, AWS Step Functions, Amazon Bedrock) Oct 2025

- Won a **Special Mention** at an **AWS hackathon** (solo) by orchestrating **3 Bedrock AI agents** over **5 live data sources** in Step Functions

IncidentLens - Autonomous AI Incident Response Copilot (FastAPI, LangGraph, Claude API, RAG) Jul 2026

- Cut on-call time-to-context from **15+ min** to **under 60 s** by building a **multi-agent LangGraph** copilot with RAG
- Hit **90%+ bad-commit detection accuracy** on **25+ simulated incidents** by tracing root causes to commits with MCP tools
- Cut postmortem drafting from **2+ hours** to **under 10 min** by auto-generating incident timelines with the Claude API